

SECTION-H DISSERTATION PREPARATION GUIDELINES AND FORMAT

This section provides detailed information regarding the preparation and formatting of the final year research dissertation. The thesis/dissertation is an essential document for this research project, and it should be in the form of a report. The thesis carries **40%** marks to the result of the final year research project. Writing a clear, concise, structured, and well-planned thesis or dissertation will capture the attention of all readers, even the readers who are not directly connected with your research topic. Students are advised to follow specified guidelines, formats, and scientific writing, etc., during the preparation of the thesis. Students must perform the initial submission of the corrected dissertation on or before the deadline, as directed by the research-coordinator. Students may submit the final hardbound thesis copies of the thesis after final presentations, on or before the deadline, as directed by the research-coordinator.

H:1 OUTLINE OF THE DISSERTATION

The final year research dissertation should be outlined as follows. Refer to the short description under each section while writing.

NOTE: - [formatting should be done as indicated in the provided specimen ****

Consult your research coordinator for additional information regarding the sample dissertation specimen or download it from the UWU website]

1)Cover Page

..... *[Roman page numbers begin here]*

2)Title Page

3)Author Declaration

4)Acknowledgment

5)Content

6)Abstract

7)List of Figures

8)List of Tables

9)List of Abbreviations

..... *[Page 1 begins here]*

10) Chapter 1: Introduction

The rationale for the research should be provided. Problem Identification, objectives of the research, and expected outcomes should be elaborated under appropriate subheadings. However, they may not be limited to these topics only.

11) Chapter 2: Literature Review

Theoretical background, previous work, and gaps in knowledge should be discussed under appropriate subheadings. However, they may not be limited to these topics only.

..... *[Both chapters together should not exceed ten pages]*

12) Chapter 3: Materials and Methods

All materials, samples, maps, and available data used in the research should be explained and should be supported with figures, tables, charts, and pictures. Analytical techniques, sampling methods, laboratory experiments, and field investigations should be explained in detail with supporting materials. Description of established, standard, lengthy analytical procedures, and data tables should be included only in the appendix.

13) Chapter 4: Results and Discussion

Results obtained should be presented and discussed in detail with direct correspondence to Chapter 3. Subsections should be made as required. A proper interpretation of the results should be made. Presentation of results in tables, charts, figures, graphs, maps, and texts should be meaningful. Irrelevant, redundant, and uninformative results should be avoided.

14) Chapter 5: Conclusions

A detailed evaluation of results and an overall assessment of the research outcomes should be presented.

..... *[Chapters 3-5 together should not exceed 40 pages]*

15) References

A list of references should be of a standard format accepted in the scientific literature. Un-cited references should not be listed.

16) Appendix

All supporting materials which are too extensive for the body text should be included here.

..... *[Page numbers end here]*

H:2 GENERAL FORMATTING

1) Paper and Printing

Each page shall be single-sided on clear white paper of good quality, having at least 80 GSM and A4 size (210 x 297 mm).

2) Margins

Leave a 3.85 cm margin on the left and 2.5 cm margins on the right-hand side, top, and bottom.

3) Font

One type of legible font (preferably Times New Roman - size 12) must be used throughout the

dissertation.

4) **Line Spacing and Paragraphs**

The line spacing should be 1.5. Keep a one-line spacing between paragraphs.

5) **Diagrams**

Drawings, diagrams, maps, etc. should be clear and may be reproduced by photographic or other processes. They should carry a unique sequence number (use appropriate according to the format specified) and captions on the same page. All figures must be cited in-text. If applicable, the source must be mentioned.

6) **Tables**

All tables should carry a unique sequence number (use appropriate according to the format specified) and captions on the same page. All Tables must be cited in-text. If applicable, the data source must be mentioned.

6) **Pagination**

All pages must be numbered, including the appendices. Place page numbers at the bottom centre of the paper. Use Arabic numbering for the main body of the dissertation and Roman numbering for the appendices and prefatory pages (Declaration to List of Tables).

7) **Guideline for Citing References with the Author-Year System**

A. **Citing References in the Main Body of Text**

- | | |
|---|--|
| i) Only one author | Eg: (Bell, 1991) |
| ii) Two authors | Eg, (Priesler & Robertson, 1992) |
| iii) Three or more authors | Eg: (Waite <i>et al.</i> , 1992) |
| iv) Author not known | Eg: (Anonymous, 1850) |
| v) Several publications by an author in the same year | Eg: (Bell, 1992a) (Bell, 1992b) vi) Two |
| references for the same text line | Eg: (Bell, 1991; Waite <i>et al.</i> , 1992) |

B. **List of References and the Bibliography**

List the references in alphabetical order according to the correct format. The formats for different types of references are as follows.

(i) **Journal article**

(a) **One author**

Bell, C.H. (1991). Diapause and cold tolerance of larvae of *Ephestia elutella*. *Postharvest Biology and Technology* 1, 81-93.

(b) Two authors

Preisler, H.K., and Robertson, J.L. (1992). Estimation of treatment efficacy when the number of test subjects is unknown. *Journal of Economic Entomology* 85, 1033-1040.

(c) Three or more authors

Waite, D.T., Grover, R., Westcott, N.D., Sommerstd, H., and Kerr, L. (1992). Pesticides in groundwater, surface water, and spring runoff in a small Saskatchewan watershed. *Environmental Toxicology and Chemistry* 11, 741-748.

(d) Several publications by an author in the same year

Bell, C.H. (1992a). Diagnosis of larvae of *Ephestia elutella*. *Postharvest Biology and Technology* 2, 8-12.
Bell, C.H. (1992b). Tolerance levels of larvae of *Ephestia elutella*. *Journal of Agriculture* 5, 81-93.

(ii) Journal article (electronic)

Williams, F. (1997). Electronic Document Delivery – a trial in an academic library. *Ariadne*, issue 10, July 15. <http://www.ariadne.ac.uk/issue10/edd/> (December 5, 1997)

(iii) Journal without volume numbers

Jang, E.B., (1991). Thermal death kinetics and heat tolerance in early and late third instars of the oriental fruit fly (*Diptera: Tephritidae*). *Journal of Economic Entomology*, 1298-1303.

(iv) Journal with new pagination in each issue

Becker, L.J. & Seligman, C. (1981). Welcome to the energy crisis. *J. Social Issue* 37 (2), 1-7.

(v) Translation journal

Assink, E.M.H. & Verloop, N. (1977). Het aanleren van dee-geheel relaties in het aanvankelijk rekenonderwijs [Teaching part-whole relations in elementary mathematics instruction]. *Pedagogische Studien* 54,130-142.

(vi) Reference to abstract of an article

Karunaratne, W.M.A.A. & Dissanayake, C.B. (1983). The distribution of goldmines and mining villages in ancient Sri Lanka. Abs. *First Geology Symposium of Sri Lanka*: University of Peradeniya.

(vii) Paper not yet published

Potting, R.P.J., Otten, H. & Vet, L.E.M. (1997). The relation between parasitoid ecology and learning: the absence of learning in the stemborer parasitoid *Cotesia flavipes*. *Animal Behaviour* (in press).

(viii) Monograph

Gunatilleke, C.V.S. (1996). *A nature guide to the world's end trail, Horton Plains*. Peradeniya Science Publication 5.

(ix) Chapter in monograph

Ugi, I. (1971). *Isonitrile chemistry*, New York, Academic Press; Chapter 2

(x) Book

Abeles, F.B. (1973). *Ethylene in Plant Biology*. Academic Press, New York.

(xi) Edited book

Worthing, C.R. & Walker, S.B. (1987). *The Pesticide Manual*, 8th edn. British Crop Protection Council, Thornton Heath.

(xii) Chapter in an edited book

Ries, S.K. (1976). Sub-toxic effects on plants. In *Herbicides: Physiology, Biochemistry, Ecology*, 2nd edn. Vol. 2, ed. L.J. Audus. Academic Press Inc. (London) Ltd, Chapter 2, 313-344.

(xiii) Edited symposia, special issues, etc., published in a periodical

Kimball, B.A. & Idso, S.B. (1983). Increasing atmospheric Carbon Dioxide: effects on crop yield, water use, and climate. In: J.F. Stone and W.O. Willis (Editors), *Symp. Plant Production and Management under Drought Conditions*, 4-16 October 1982, Tulsa, OK. *Agric. Water Manage.* 7, 55-72.

(xiv) Patent

Kysika, J.O., Sawicki, C.A. (1983). Apparatus and method for measuring optically active materials, U.S. Patent 352 321.

(xv) Thesis

Cregg, B.M. (1990). Net Photosynthesis and carbon allocation of loblolly pine (*Pinus taeda L.*) branches in relation to three levels of shade. Ph.D. thesis, University of Georgia, USA.

(xvi) Thesis (website)

Smith, J. (2000). *Curly's Airships Polegate, Masters of Arts* Available from <http://www.curlysairships.com> (Accessed 29 May 2001).

8) Guideline for Citing References with the Number System

A. Citing References in the Main Body of Text

Use inline square-bracketed numbers for each reference.

- i) Only one reference Eg: [1]
- ii) Two references for the same text line
 Eg: [1,2]
- iii) More reference for the same text line
 Eg: [3, 6-9]

B. List of References and the Bibliography

In the Numeric System references are listed in the order they appear in the text.

Eg:

- [1] Nakamishi, T., Ito, K., *Sol. Energy. Mat.* 1994, 35, 171.
- [2] Williams, F. Electronic Document Delivery – a trial in an academic library. Ariadne issue 10, 15 July 1997. <http://www.ariadne.ac.uk/issue10/edd/> (5 December 1997)
- [3] Hart, T.W., *J. Chem. Soc. Chem. Commun.* 1979,156.
- [4] Haggin, I., *Chem. Eng. News* 1985, 63 (42), 23-25.
- [5] Volpin, M.E., *J. Gen. Chem. USSR (Engl. Transl.)* 1960, 30 1207; *Zh.Obschch. Khim.* 1960, 30, 1187.

C. Recommended Format for Numeric System

(i) Journal article

- [1] Nakamishi, T., Ito, K., *Sol. Energy. Mat.* 1994, 35, 171.

(ii) Journal article (electronic)

- [2] Williams, F. Electronic Document Delivery – a trial in an academic library. Ariadne issue 10, 15 July 1997. <http://www.ariadne.ac.uk/issue10/edd/> (5 December 1997)

(iii) Journal without volume numbers

- [3] Hart, T.W., *J. Chem. Soc. Chem. Commun.* 1979,156.

(iv) Journal with new pagination in each issue

- [4] Haggin, I., *Chem. Eng. News* 1985, 63 (42), 23-25.

(v) Translation journal

- [6] Volpin, M.E., *J. Gen. Chem. USSR (Engl. Transl.)* 1960, 30 1207; *Zh.Obschch. Khim.* 1960, 30, 1187.
- (Translation journal data followed by original journal data)

(vi) Reference to abstract of an article

- [6] Mirnov, V.F., *Izv. Akad. Nauk SSSR* 1966,1177; *Chem. Abstr.*1966, 65,16997.

(vii) Paper not yet published

- [7] Ariyaratne, K.A.N.S., *J. Natl. Sci. Coun. Sri Lanka*, in press.

(viii) Monograph

- [8] Soo, S.L., *Fluid dynamics of Multiphase systems*. NewYork: Blaisdell, 1967.

(ix) Chapter in a monograph

- [9] Ugi, I., *Isonitrile chemistry*, New York: Academic Press, 1971; Chapter 2.

(x) Book

- [10] Smith, J.D., Crawford, F., *Hormonal Mediators in Bees*. Greenville: The New Science Press, 1985.

(xi) Edited Book

- [11] Clearfield, A. (Ed.) *Inorganic Ion Exchange Materials*. Boca Raton: CRC Press, 1982.

(xii) Chapter in an edited book

[12] Stoeppler, M. and Nireuberg, W., in: *Metalle in der umwelt*; Merian, E. (Ed.) Wlinheim: Verlag Chemie, 1984; Chapter I, 4a.

DISSERTATION EVALUATION CRITERIA

The **Appendix-G4** exhibit the thesis evaluation criteria for the final year research project. This document is addressed to supervisors and examiners. It provides information on general considerations and assessment criteria for the project. It also guides students about what examiners will be looking for in examining the projects. The supervisor must submit the completed form “**Appendix-G4**” to the research coordinator of the corresponding department, Faculty of Applied Sciences, Uva Wellassa University, on or before the deadlines as specified by the research coordinator. Mark allocation for evaluation of the final year research project at different stages and their contribution to the final grade is summarized in the “**Appendix-G5**” as a reference for both students and examiners.

GENERAL EVALUATION GUIDELINES

The examiners may consider following during the evaluation of the final year research project thesis, but not limited to these facts.

- Are the project aims (including the research problem, research questions, boundaries, and desired outcomes) well formulated?
- Are the background conditions and motivation described in enough detail to provide a rationale for the project?
- Are relevant concepts and empirical studies in open literature critically reviewed?
- Is the methodology fit with aims and the stated problems of the study appropriately?
- Is the methodology consistent and reflecting an adequate amount of effort for an undergraduate student?
- Are the findings and experiences well summarized?
- Are the findings well discussed, and are the implications valid for the candidate’s situation and conditions?
- Is it an original work of the student, and what is the contribution to the area of research?
- Does the student maintain minimum expectations from scientific writing, including the style of writing, presentation, and grammatical expression?
- Is the study potential to publish in peer-reviewed journals or conferences?